

Phase 3 Abstract

1. Project Overview and Concept

RevUp Rental is a motorcycle rental web application created for people who are motorcycle enthusiasts, tourists, commuters, and adventure travelers. The main idea behind the application is to provide a seamless, user-friendly platform where customers can browse a selection of motorcycles, explore available accessories, and complete a rental booking from a single website. Users benefit from seamless booking, delivery, and pickup services, 24/7 support, and a wide selection of bikes ranging from daily riding scooters to premium models such as the Ducati Diavel and BMW R1250 GS Adventure. The project was approached during the conception phase with a clear three-tier architecture in mind, a frontend built with HTML5, CSS3, and JavaScript, and a backend powered by PHP running on a XAMPP local server, and a MySQL relational database for structured data storage. This architecture was chosen because all three components are industry-standard, well-documented, and well-suited for the basic full-stack web project. PHP and MySQL integrate tightly and offer a low-barrier path to implementing server-side logic and persistent data management without requiring a separate API layer.

2. Technical Approach and Implementation

The frontend code includes a single-page-style HTML template divided into logical webpage sections (Home, About, Motorcycles, Gallery, Reviews, Accessories), navigable through a fixed navbar at the top. Cascading Style Sheets (CSS3) was used to style every element on the page and create hover animations, card stylizations, responsive page elements, and smooth scroll behavior. JavaScript is used for two features on our page, they are an alert that activates when a non-signed-in user attempts to click on one of the rental buttons, and form validation on sign-up and booking forms. PHP serves three specific purposes on our page, user creation through signup, authentication of login information, and form submission for bike rentals. Database PHP scripts use parameterized MySQL queries to access and insert data into the relevant MySQL database. Relations were created between our four main database tables, such as users, motorcycles, rentals, and accessories, to prevent duplicate information. For instance, our rentals table had to include FK constraints for users.id and motorcycles.id to track which user rented which motorcycle and when. One interesting component of our code was the use of PHP to create sessions upon login. When a user successfully logs in, their username and password are placed into a PHP session. The logout button on the navbar was written in PHP to only be displayed if a user is logged in. Lastly, the book ride form autofills the username using PHP session data.

3. Changes from Conception to Implementation

Minor differences between Phase 1 Planning and Phase 2 completion include additional detail added to components that were originally kept very broad. The main addition was the complete User Authentication system. During phase 1, only the logic behind how the approach to login and signups

was described but never went into detail about what the PHP would look like. As coding began, it was decided that it was necessary to have users create an account and log in to the website to prevent people from easily manipulating our rental data. What was previously described as a data component in phase 1 became the MySQL relational database. Version control via GitHub and development in Visual Studio Code were also introduced during Phase 2.

4. Responsive Design

A key requirement was to make sure the frontend is readable across different viewport sizes. The CSS was structured with elements, percentage-based widths where possible, and media queries targeting mobile and tablet breakpoints. The navigation bar adapts smoothly on smaller screens, the motorcycle card grid reduces from four columns to two and then one, and the gallery grid adapts from three columns to a single-column layout on mobile. These changes ensure the site remains fully usable on smartphones and tablets without altering any of the underlying HTML content or structure, and to avoid horizontal scrolls.

5. Test Cases and Validation

Testing was carried out manually across key user flows. Representative test cases included, firstly, clicking "Rent Now" without logging in, the expected result is that an alert pop-up appears, and the user is redirected to the signup page. Second is submitting the signup form with an already-registered email, the expected result is that an error message is displayed, and no duplicate record is created. Third is submitting the booking form with all valid fields while logged in, the expected result is that the booking is saved to the database, and a confirmation is shown. Fourth is resizing the browser window to any device's viewport, the expected result is that the navbar stacks vertically and all content remains readable without horizontal scrolling. All test cases produced the expected results.

6. Lessons Learned

The most important lesson from this project was the importance of planning the database schema carefully before creating the database. Going forward, an entity-relationship diagram was created before coding began to prevent errors. A second lesson concerned responsive CSS, the initial stylesheet relied heavily on fixed pixel values, which caused layout breakage at different screen sizes. Refactoring these into relative units and media queries was time-consuming but produced a much more robust and responsive frontend. Finally, working with PHP sessions and MySQL on a local XAMPP server highlighted the importance of environment consistency. Minor version differences between PHP and MySQL occasionally caused unexpected query behavior. Documenting the exact version stack in the README.md from the beginning would have saved debugging time and made the project easier to reproduce on a different machine. Overall, this project strengthened both technical skills and systematic development practices, such as planning, version control, testing, and documentation, as equal contributors to a successful web application.